# DYNAMIC RESTAURANT QUEUE MANAGEMENT SYSTEM

Submitted in partial fulfillment of the requirements of the degree of

## BACHELOR OF ENGINEERING

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2025-2026

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# CERTIFICATE

This is to certify that the project entitled "Dynamic Restaurant Queue Management System" is a bonafide work of "**Raj Patil" (22106135), "Anup Singh" (22106104), "Yogesh Vartak" (22106119), "Soham Waradkar" (23206002)** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering** in **Computer Science and Engineering (Artificial Intelligence and Machine Learning).**

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# Project Report Approval for T.E.

This project report entitled ***Dynamic Restaurant Queue Management System*** by **Raj Patil, Anup Singh, Yogesh Vartak, Soham Waradkar** is approved for the degree of ***Bachelor of Engineering*** in **Computer Science and Engineering (Artificial Intelligence & Machine Learning),** ***2025-26***.

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## Declaration

We declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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# Abstract

During peak hours, college canteens often face heavy congestion and long queues, resulting in significant waiting times for students, faculty, and staff. This problem not only wastes valuable time but also reduces the overall dining experience, leading to dissatisfaction among both customers and canteen staff. The proposed project to address these issues by developing a comprehensive digital platform that integrates customer ordering, kitchen workflow management, and administrative control.

The system will provide a customer-facing mobile and web application that allows users to browse menus, customize items, schedule orders, and make payments through Unified Payments Interface (UPI) or cash on pickup. Customers will receive real-time updates about their orders, including estimated preparation times (ETAs), status notifications such as "Accepted," "Preparing," and "Ready for Pickup," and QR codes for smooth verification at the counter. Features such as favorite combos, order history, and quick reordering will further enhance convenience and encourage repeated usage.

On the kitchen side, the project introduces a Kitchen Display System (KDS) that replaces traditional paper tickets. The KDS displays all incoming orders in a structured, color-coded, and prioritized queue. It enables batch preparation of similar items, highlights delayed or scheduled orders, and routes items to appropriate stations such as beverages, fried food, or Chinese cuisine. Staff can update order statuses with a single tap, and automated alerts will notify them of delayed or overdue items, ensuring faster and more accurate service.

The Admin Dashboard forms the third major component of the system. It provides canteen managers with tools to manage the menu, monitor inventory, mark items as out of stock, and analyze performance using key performance indicators (KPIs) such as busiest hours, most ordered items, and average preparation time. This ensures that administrators can make datadriven decisions, optimize staffing, and improve overall efficiency.

By combining these three perspectives—customer, kitchen staff, and administrators—Smart Canteen creates a holistic solution to minimize queues, reduce preparation errors, and improve service delivery. The system aims to leverage digital technologies and machine learning–based ETA prediction to offer a seamless dining experience. Ultimately, this project demonstrates how the integration of pre-ordering, KDS, and analytics can transform the operation of a college canteen into a more efficient, transparent, and user-friendly ecosystem.

The project contributes to several Sustainable Development Goals (SDGs). It supports Industry, Innovation & Infrastructure by introducing smart technology to enhance food service infrastructure and enable digital queue management. It aligns with Sustainable Cities & Communities by reducing crowding and improving efficiency in public eating spaces such as canteens and food courts. Additionally, it promotes Responsible Consumption & Production by minimizing food waste and overproduction through synchronized food preparation based on customer arrivals.

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**ABBREVIATION**

KDS – Kitchen Display System
AI–Artificial Intelligent
ETA – Estimated Time of Arrival
UPI – Unified Payments Interface
COD – Cash on Delivery
DB – Database
ML – Machine Learning
JWT – JSON Web Token
API – Application Programming Interface
REST – Representational State Transfer
SQL – Structured Query Language
NoSQL – Not Only SQL
SMS – Short Message Service
UAT – User Acceptance Testing
VM – Virtual Machine
KPIs – Key Performance Indicators
QSR – Quick Service Restaurant
PG – Payment Gateway

# Chapter 1

# Introduction

## 1. Introduction

The college canteen is one of the busiest places on campus, especially during lunch breaks and peak hours. With limited staff and a large number of students, faculty, and staff relying on it, the canteen often faces issues such as overcrowding, long queues, and delays in food preparation. These problems not only waste valuable time but also create stress for both customers and kitchen staff.

In recent years, digital transformation in food service management has shown promising results in reducing these challenges. By integrating mobile applications, online ordering systems, and real-time kitchen management tools, it is possible to optimize service delivery and enhance customer satisfaction. This project is focused on applying such solutions to the college canteen environment to save time, reduce waiting lines, and improve overall efficiency.

## 1.1 Background & Motivation

College canteens typically face two problems: (a) concentrated demand during short lunch/break windows and (b) inefficient coordination between order taking and kitchen preparation. Long queues result in lost time for students and staff and inefficient utilization of kitchen capacity. Pre-ordering and a digital kitchen interface have emerged as effective solutions to reduce waiting times and errors, as shown in multiple deployments and studies.

## 1.2 Aim of the Project

To design, implement and evaluate a full-stack application that enables campus users to preorder and schedule canteen food remotely, route orders to kitchen staff via a KDS, provide real-time status & ETA, allows admin management (menu, inventory, analytics) and supports secure payments.

## 1.3 Scope

- Customer app: menu browse, search, customization, schedule, payment, notifications.
- Kitchen staff KDS: real-time queue, category routing, timers/alerts, status updates.
- Admin dashboard: menu management, inventory sync, analytics, manage staff and offers.
- Backend APIs, DB, authentication & logging; simple ML model to predict item preparation time.

The proposed application aims not only to simplify the process of ordering food but also to bring together customers, kitchen staff, and administrators on a single digital platform. By integrating these three components, the system will minimize miscommunication, speed up service delivery, and ensure that resources in the canteen are used effectively.

Furthermore, the inclusion of predictive features such as estimated preparation times and order scheduling will improve transparency and allow customers to plan their visits in advance. This reduces unnecessary waiting, particularly during peak hours, and enhances overall user satisfaction. From the perspective of kitchen staff, the Kitchen Display System (KDS) will help streamline food preparation by displaying incoming orders in real time, grouping similar items, and highlighting scheduled or delayed requests. This structured approach will reduce human errors, reduce wastage of food and ensure that the promised preparation time is met.

# Chapter 2

# Literature Survey of Existing systems

A literature survey is a key component of any research project, giving us a glimpse into existing solutions, pinpointing gaps in current methods, and highlighting the significance of the proposed system. When it comes to online food ordering and managing canteens, there's been quite a bit of research and implementation at both academic and industrial levels. These efforts lay the groundwork for creating a solid solution that combines customer apps, Kitchen Display Systems, and admin dashboards.

## 2.1 Pre-ordering Systems for Campus Canteens

One of the first strategies to boost campus canteen efficiency was introducing pre-ordering systems. These systems let students and staff order their food ahead of time, cutting down the wait in line. Research indicates that this leads to a smoother order process and better time management during busy hours. For example, a study titled Pre-ordering Mobile Application for Campus (2022) showed that students could schedule their meals in advance and simply pick them up without wasting time in long queues. This approach eased congestion during break times and improved the dining experience. Similar initiatives have been tested in various universities, consistently showing better service efficiency.

## 2.2 Comprehensive Surveys on College Canteen Efficiency

In 2024, a survey called Enhancing College Canteen Efficiency through Online Food Ordering: A Comprehensive Survey looked at how online ordering platforms have been embraced across multiple institutions. The findings showed that digital ordering can help manage the rush during peak hours, boost customer satisfaction, and provide canteen operators with better sales tracking. However, it also pointed out some drawbacks, like lacking real-time updates with kitchen staff and limited analytics for admins. These shortcomings underline the need for a more integrated solution that brings together customer ordering, kitchen workflow management, and administrative oversight.

## 2.3 Kitchen Display Systems

Kitchen Display Systems have been extensively researched in the restaurant sector as a replacement for traditional paper tickets. Studies and industry reports consistently point out KDS benefits, such as reducing order mistakes, shortening delays, and enhancing communication between front-end and back-end teams. For instance, case studies show that restaurants using KDS have achieved up to a 40% drop in average order fulfilment times. KDS also offers features like color-coded order statuses, batch preparation grouping, and prioritization of delayed or scheduled orders. Even though these systems are popular in commercial restaurants, their use in campus canteens isn't widespread yet. This presents a chance to adapt and implement KDS specifically for academic settings, leading to better coordination among staff and a more enjoyable experience for students.

## 2.4 Mobile Order-Ahead Applications

Mobile order-ahead apps like those from Starbucks and McDonald’s have been evaluated for their impact on managing queues. Robbins (2021) examined how these apps affect quick service restaurants (QSRs) and found they significantly cut down on physical lines by pushing demand to digital platforms. This allowed kitchens to start preparing items before customers arrived, increasing throughput and enhancing customer satisfaction. The same idea could work in college canteens, where short breaks and high demand often create bottlenecks.

## 2.5Literature Review

1. G. M. Aditya, A. Hoode, A. Rai K, G. Biradar, A. Kumara M. A, M. Kumar "ML-Based Platform and Recommendation System for Food Ordering Services Within Premises," Proceedings of the International Conference on Computer Science and Engineering, pp. 129–137, 2025. This study suggests a machine learning-based in-premise food ordering platform that features personalized meal recommendations and operational optimization. Through collaborative filtering and clustering, it offers customized meal suggestions that adapt to user preferences in real time. The analytics help optimize kitchen workflows and inventory management, ultimately improving user experiences and reducing wait times.

2. S. Biradar, N. Chavan, R. Gaikwad, P. Godase, B. D. Phulpagar, "Intellicater - ML Powered Canteen Management System," International Journal of Research Publication and Reviews, vol. 5, no. 5, pp. 71-81, 2025. This research creates a canteen management system enhanced with machine learning techniques for personalized menu recommendations based on customer feedback and purchase history. Intellicater utilizes predictive analytics to dynamically adjust the menu according to customer preferences, optimizing resource allocation. It automates menu management, decreasing the manual labor involved in planning, and streamlines dining room operations, leading to better customer satisfaction and operational efficiency.

3. Anwitha, Dhanushree BA, Monika CV, Darshini MS, "Smart Canteen System Using AI, Real-Time Analytics, and Cashless Integration," International Journal of Novel Research and Development, vol. 4, no. 6, pp. 842–850, 2025. This paper introduces a digital canteen system that automates order management, cashless payments, and real-time inventory tracking. AIdriven analytics predict demand and track trends, helping minimize food wastage and improve customer service through quicker order fulfillment and transparency. The system incorporates IoT sensors and data analytics for cost-effective and efficient food service management.

4. Tejasri, D. Vinay Kumar, G. Prasad, K. Deeksha, CH Devi, S. Krishna Deepika, "Intelligent Canteen Management Using Machine Learning for Real-Time Ordering and Sustainable Operations," International Journal of Innovative Research in Technology, vol. 11, no. 11, pp. 5295–5301, 2025. This paper tackles common challenges faced in campus canteens like long lines, missed meals, and food waste leading to dissatisfaction. Traditional inventory management methods often lead to inefficiencies and human errors that result in either food shortages or surpluses. The proposed system combines real-time data processing with machine learning algorithms for demand forecasting using the Prophet model and order pattern analysis with K-Means clustering. It offers a digital platform for viewing daily menus, pre-ordering, instant notifications, and digital payments, helping reduce wait times and manage bulk orders better. The system automates inventory control, tracks orders, processes payments, and includes a chatbot for improved customer interactions, enhancing

operational efficiency, reducing waste, cutting costs, and boosting sustainability while improving customer satisfaction overall.

5. P. Bari, Y. Patil, N. Sathawne, T. Suroshe, and A. Patil, "Deploying an Intelligent Online Food Ordering System to Optimize College Canteen Operations," Proceedings of the 3rd International Conference on Futuristic Technology (INCOFT), vol. 3, pp. 263–268, 2025. This study introduces a web and mobile platform built on the MERN stack designed to streamline college canteen ordering through real-time processing and machine learning for personalized meal recommendations. The system cuts down wait times, improves order accuracy, and enhances user experience by supporting digital payments, queue management, and role-based access. Machine learning assists in predicting user preferences and inventory needs, optimizing kitchen workflows and inventory management to boost operational efficiency and overall campus dining satisfaction.

# Chapter 3

# Limitation of Existing system

## 3.1 Common limitations

Even though there are quite a few online food ordering and canteen management systems out there, many of them come with some pretty significant drawbacks.

For starters, a lot of apps aimed at college students are just standalone tools for placing orders. They let students make their choices, but they don't connect with the kitchen. So, without a Kitchen Display System (KDS), the staff often have to rely on paper tickets or shout out orders, which can lead to delays and mix-ups.

Another issue is that they usually lack robust analytics and inventory tracking features. This means administrators can't easily see things like peak ordering times, what dishes are the most popular, or stop customers from ordering things that are out of stock. This definitely hurts both the efficiency of operations and overall customer satisfaction.

Most of these systems also have a problem with how they handle estimated preparation times (ETAs). Either those times are set in stone or adjusted manually, which doesn't account for things like how complicated an order is or how busy the kitchen is. This often leads to inaccurate delivery times. On top of that, existing systems usually don't offer much in the way of flexible prioritization for orders, whether they're scheduled or running late, so kitchens struggle to deliver on time.

There are some solutions that dabble in virtual tokens or QR codes, but these features often don't sync up well with order tracking and kitchen processes. So, it's clear there's a big gap here: we really need a unified platform that brings together customer ordering, kitchen integration, predictive ETAs, inventory synchronization, and administrative analytics.

# Chapter 4

# Problem Statement and Objectives

## Problem Statement & Objectives

### 4.1 Problem Statement

Creating a Smart Canteen Management System that aims to solve issues like long queues, manual order processing, inventory mismanagement, and food wastage in college canteens by integrating pre ordering, real-time order tracking, AI-powered demand forecasting, personalized menu recommendations, and cashless payment options to provide a seamless, efficient, and user-friendly food service experience for students, staff, and canteen administrators.

### 4.2 Objectives

- To develop a mobile app that allows students to pre-order meals, customize items, and schedule pick-up, with digital ordering capabilities.
- To implement a Kitchen Display System that receives orders from the app and displays preparation status, ensuring seamless communication between front-end and kitchen staff.
- To employ predictive analytics on historical and live kitchen data to estimate order preparation and arrival times, providing accurate ETAs to users.
- To reduce in-person queue lengths and student wait times by coordinating order schedules and pickup slots.
- To enhance overall canteen efficiency and food service quality, including waste reduction and higher student satisfaction, through data-driven optimizations.

# Chapter 5

# Proposed System

## 5. Proposed System

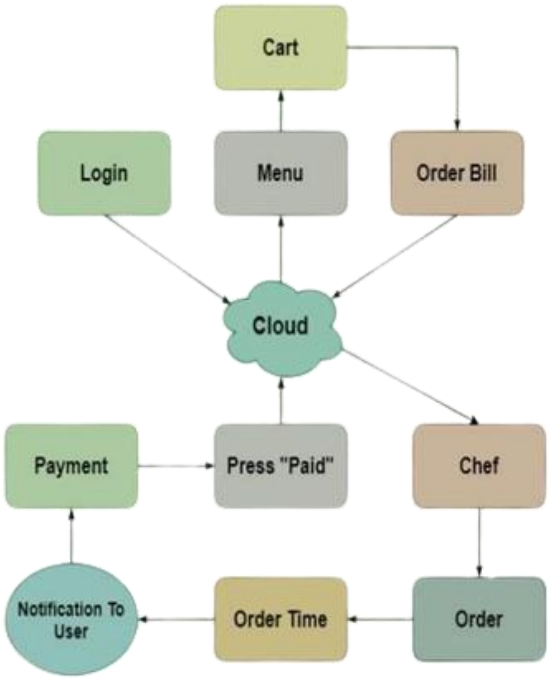
### 5.1 Introduction and Strategic Methodology

In this chapter, we dive into the technical setup, design details, and method that drives the Cafeteria Helpdesk system. Our main aim is to create a solution for the uncertainty around timing by setting up real-time, Just-in-Time (JIT) pickup coordination. We're going with the Hybrid Control System (HCS) method, which brings together predictive automation and human oversight to handle the unpredictable nature of the canteen.

The setup is built on a Mobile-First, Serverless Backend-as-a-Service (BaaS) model. We’re using Flutter for quick cross-platform client development and Firebase to ensure smooth, lowlatency data syncing and secure processing of our business logic with Cloud Functions. This setup not only guarantees resilience but also meets our budget and maintenance requirements.

### 5.2 System Architecture and Framework Analysis

The Cafeteria Helpdesk features a three-tier logical architecture meant for high activity, simultaneous operations, and minimal delays—essential for a system that needs to manage busy periods in real time.




### 5.2.1 High-Level Architecture (Client-BaaS-Client Model)

This architecture follows a modern Client-BaaS-Client model, where Google Firebase acts as the central hub. This design takes away the headaches of infrastructure management, letting us focus purely on core functionality and security.

- **Client Layer (Flutter):** Built with Flutter, this layer offers a cross-platform mobile interface that lets users browse menus, place orders, track ETAs, and get real-time updates without a hitch.
- **Student App:** Connects with Firebase Auth for user verification and uses Cloud Firestore to keep tabs on order statuses. When users place an order, it goes directly to Firestore.
- **Kitchen Staff App:** Works with Firebase Auth as well, checking the current order queue from Firestore and executing commands (like ETA changes or status updates) through secure Firebase Callable Cloud Functions.
- **BaaS Layer (Firebase Core):** Handles data storage, user management, file storage, and notifications.
- **Serverless Logic Layer (Cloud Functions):** This part manages all important business logic, security checks, and heavy calculations in a safe, trusted environment.

### 5.2.2 Technology Framework Justification

| Technology | Architectural Role | Technical Justification |
|---|---|---|
| Flutter/Dart | Native-Client Development | Enables code reuse and consistency across the Student and Staff apps |
| Cloud Fire-store | Real-Time Synchronization Engine | Selected over the Realtime Database for its superior document query capabilities necessary for filtering the KDS queue. |
| Cloud Functions | Transactional-Logic Gateway | Enforces business rules and performs atomic updates securely. |

## 5.3 Detailed Data Model and Security Implementation

The system's effectiveness hinges on a solid data model and top-notch security measures.

### 5.3.1 Cloud Firestore Data Model Design

We use denormalization in the primary document to make sure that all the info needed for a single order transaction is accessible in just one read/write operation. This helps reduce mobile latency and keeps read costs low.

## Key Technical Fields and Their Purpose:

**finalETA:** This is the key field that acts as the sole source for the countdown timer seen on a student's phone.

**Audit Array:** Here's a non-changeable array that keeps track of every major lifecycle event – like status changes, ETA adjustments, timestamps, and the ID of the person making the changes. This is vital for thorough oversight and accountability.

**idempotenceKey:** A GUID generated by the client. If this key pops up in an order that's just been completed, the function halts to prevent duplicate orders from being made during mobile network retries.

### 5.3.2 Firestore Security Rules (RBAC Implementation:

We handle security at the database level using Role-Based Access Control (RBAC), backed by custom claims in Firebase Authentication tokens.

**Student Access:**

**For creating records:** Needs authentication and checks that the request.resource.data.userId matches the authenticated request.auth.uid.

**For reading records:** Only granted if it matches the auth.uid**.**

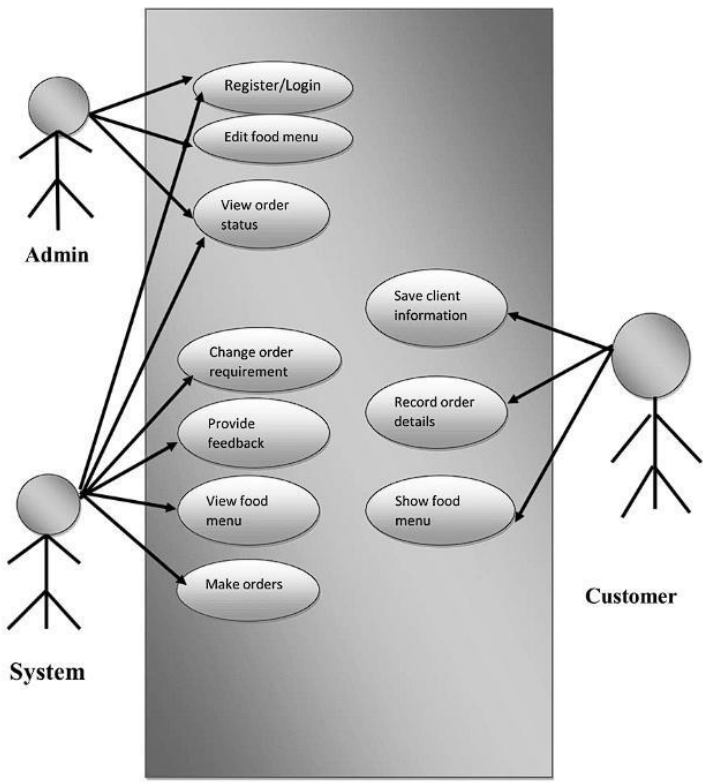
**For reading and updating:** Requires the user to have the role claim (request.auth.token.role 'kitchen').

**Field-Level Validation:** Staff cannot directly change the fields. They must update through the field and launch the via the Callable Cloud Function, which has its own trusted validation process.

### 5.4 Core Operational Flows and Hybrid Control Logic

The system's intelligence operates through three essential workflows, mainly coordinated by Cloud Functions, showcasing the Hybrid Control System (HCS) in action.

#### 5.4.1 Flow A: Order Placement and Auto ETA Initialization Flowchart




**5.4.1 Use Case Diagram for Online Canteen Ordering System**

**Process Steps and Serverless Logic:**

- **Client Write:** The Student App writes a document marking the status as PLACED.
- **CF Trigger:** The Cloud Function kicks in immediately after the document is created.

- **Idempotency & Settings Check:** The Cloud Function reads the and pulls settings (maxParallelOrders, baseBufferMinutes).
- **Atomic Write:** The function sets autoETA, makes finalETA equal to autoETA, and
- generates the pickup code.
- **FCM Alert:** FCM sends "Order Accepted" notification. The Staff App's listener updates the KDS queue.

### 5.4.2 Flow B: Hybrid Control - Manual ETA Override Flowchart

This is the most critical flow, where the system's prediction is refined by authorized staff input, implementing the HCS principle.

**Process Steps and Serverless Logic:**

- **Staff Action (KDS-as-Mobile):** Staff taps the button for an order.
- **CF ( ) Call:** Staff App invokes the secure **Firebase Callable Function**.
- **CF Logic Execution (Trusted):**
- **Authorization Check:** Verify staff role and permissions.
- **Constraint Validation:** Verify is within the array (e.g., [10, 15, 20, 30])
- **Atomic Transaction:** CF performs a transaction to simultaneously set , update the analytics fields, and append the action to the array.
- **FCM Push:** Student receives the "Updated ETA" notification. Student's timer is instantly synchronized.

## 5.5 Detailed Auto ETA Calculation Algorithm (Foundation of HCS)

The initial algorithm is designed to be highly predictive by modeling both intrinsic item complexity and extrinsic queue contention. This logic runs securely within the Cloud Function.

### 5.5.1 Algorithm Rationale and Input Parameters

The algorithm must not merely add up prep times, but account for parallel processing limitations.

- **Intrinsic Complexity:** Defined by the longest preparation time among all items in the order, as the order cannot be ready until its most time-consuming item is finished.
- **Extrinsic Contention:** Defined by the time needed to clear the current bottleneck, determined by comparing against the kitchen's configuration.

### 5.5.2 The Hybrid ETA Mathematical Formula

The final is calculated based on the maximum of the time required to complete the item itself, or the time required to get behind the current slowest order in the queue.

- **Determine Preparation Bottleneck ():** This step identifies which part of the kitchen process is the slowest or most time-consuming (for example, frying or baking). The bottleneck is the step that limits how fast an entire order can be completed.
- **Estimate Queue Waiting Time ():** This component calculates the time until the last currently active order is predicted to finish.
- **If (Contention):** Contention means multiple orders require the same kitchen resource at the same time (e.g., two orders both need the fryer). In this case, the model adds extra waiting time to account for shared equipment usage and slower overall processing.

# Chapter 6

# Experimental Setup

## 6.1 Introduction and Rationale for Experimental Design

In this chapter, we take a close look at the environment, resources, and validation methods that were set up for the development, deployment, and testing of the Cafeteria Helpdesk system. Designing the experimental setup is really important because it needs to show that our Hybrid Control System (HCS)—which is the main innovation—works better than traditional service models when it comes to Just-in-Time (JIT) Pickup Coordination.

We break down the setup into three key parts:

**Hardware Requirement**

- Processor: Intel i5 or equivalent minimum; recommended i5 for smoother builds and ML training.
- RAM: Minimum 8GB; recommended 16GB or more for efficient multitasking.
- Storage: At least 256GB SSD for faster read/write during development.
- Mobile Devices/Emulators: Android device or emulator supporting Flutter SDK (Android 8.0+ recommended).
- Network: Reliable high-speed internet for Firebase and API communication.

**Software Requirement**

- Flutter SDK: Latest stable version for cross-platform mobile app development.
- Dart SDK: Compatible with Flutter SDK.
- Android Studio/Xcode: For Android and iOS builds/emulator setup.
- Python 3.8+: For backend ML API development.
- FastAPI or Flask: Lightweight web frameworks for building RESTful APIs.

**Input Data Modeling and Simulation Protocol**: This part covers how we define the configuration parameters and the approach we use to simulate high-density order traffic to really test the algorithm under pressure.

**Performance Evaluation Parameters**: Lastly, we set out the exact metrics we'll use to measure the system's success, especially focusing on how much we've managed to reduce the time students spend waiting to pick up their orders.

### 6.2 Software and Hardware Environment (The Technology Stack)

The project utilizes a modern, serverless technology stack optimized for rapid development, automatic scalability, and high-fidelity real-time communication.

| Component | Technology / Platform | Configuration Details | Rationale for Selection |
|---|---|---|---|
| Client Framework | Flutter SDK | Stable Channel (latest) | Enables native, singlecodebase development for both the iOS/Android Student App and the KDS Staff App, minimizing complexity and maximizing code reuse. |
| Backend-as-a-Service (BaaS) | Firebase Suite | Cloud Firestore , Authentication, FCM, Cloud Functions | Provides essential real-time synchronization, secure rolebased access control, and low-latency push notifications without requiring manual server configuration or scaling management. |

| | | | |
|---|---|---|---|
| Serverless Runtime | Node.js / TypeScript | LTS Version | Chosen for the development of Cloud Functions, offering a high-performance, asynchronous environment necessary for complex, I/Oheavy tasks like the Auto ETA calculation. |
| Version Control | Git / GitHub | Standard Protocol | Ensures collaborative development |

### 6.2.2 Hardware Deployment and Validation Setup

The choice of hardware supports our idea that we can use standard, affordable mobile devices for advanced queue management.

**Development and Simulation Setup:** We work with powerful desktop computers that run the IDE, local emulators, and load-testing scripts to mimic peak-hour order arrival rates.

**Student Client Simulators (Demand Side):** Testing involves running multiple Android AVDs (Android Virtual Devices) and iOS Simulators at the same time. This helps us see how Firestore handles high demand when lots of students are subscribing concurrently.

**Kitchen Staff Console (KDS-as-Mobile):** The Staff App is set up on a specific 10-inch Android Tablet or similar mobile device. This checks how well the staff interface works in a typical kitchen, confirming that the touch controls for overriding ETAs and changing statuses are effective without the need for fixed hardware.

**Server Environment:** Our production system runs entirely on Google Cloud Platform using Firebase. This serverless setup takes care of resource management and scaling on its own, which means it can easily manage unexpected traffic spikes, like during lunchtime rushes, without needing manual adjustments or risking service quality.

## 6.3 Input Data Modelling and Simulation Protocol

To provide meaningful validation, the system must be tested not just for functionality, but for its resilience and accuracy under stress.

### 6.3.1 Static Configuration Data (Algorithmic Inputs)

The initial system configuration acts as the input for the Auto ETA Calculation Algorithm and the Hybrid Control interface.

| Parameter | Configuration value | Importance to HCS Validation |
|---|---|---|
| Menu Items and Item Preparation Minute | 40 unique items with times ranging from 2 mins to 15mins | Validates Non homogenous service time modelling: Ensures the maxTimePrep component of the algorithm works distinguishing simple vs complex orders |
| Max ParallelOrders | Set to 40-50 orders | Used to determine when the queue Wait component of the Auto ETA must activate to prevent over-promising |

### 6.3.2 Simulation Protocol and Dynamic Traffic Modeling

To really compare results effectively, we need a solid simulation protocol in place.

**Baseline Scenario (Control Group):** Here, we run the system using a simple FIFO (First-In, First-Out) model, which mimics the manual systems we currently have. Students will only see a binary 'Ready' status, which means they'll be stuck waiting at the pickup counter after placing their orders. We'll record the Average Pickup Queue Time as the baseline.

**HCS Scenario (Experimental Group):** This time, we activate the full Hybrid Control System, giving students a more interactive experience.

**Traffic Simulation:** We'll create synthetic orders that simulate peak-hour traffic, generating orders at a rate that mimics busy times (like 2 orders every second for 5 minutes).

**Stress Test Phase:** During this simulation, a test operator, acting like kitchen staff, will manually adjust ETAs to create unexpected delays (like pushing a 10-minute order back by 5 minutes). This is to see how well the system can quickly adapt to changes.

**Data Capture:** The system will automatically log important timestamps for each order, which we'll use to validate our findings:

- Time of Order Placement
- Time when final ETA was Set/Updated
- Time Status changed to READY
- Time of Actual Pickup

### 6.4 Performance Evaluation Parameters (Success Criteria

We'll determine the project's success based on a strict set of quantitative performance metrics to ensure the engineering solution meets the goals laid out in Chapter 4.

#### 6.4.1 Primary Success Metric: Service Coordination Efficiency

**Average Student Pickup Queue Time (Reduction Percentage):** This is the key measure, directly reflecting the value of the JIT coordination system and how much time it's saving students.

For the project to be considered successful, we need to see a statistically significant reduction (Target: ) when comparing the HCS Scenario with the Baseline FIFO Scenario.

#### 6.4.2 Secondary Metrics: Predictive Accuracy and System Reliability

**Predictive Accuracy (On-Time Pickup Percentage):** This measures how accurate the system is with pick up times (whether it's set automatically or manually overridden).

On-Time Pickup % = Orders Picked Up within ±3 minutes of final ETA / Total Orders × 100

We’re aiming for a target of over 90% to ensure the system is reliable in its time communication.

**Real-Time Fidelity (Latency Measurement):** This looks at how responsive our serverless architecture is

We’ll measure the delay between when the Cloud Function finishes a write operation (like setting the status to READY) and when the Student App gets the Firestore Snapshot update.

Our target Latency needs to be in milliseconds to guarantee the JIT alert system is quick and reliable.

**Hybrid Control Validation (Override Analysis):** This metric will help inform the development of future machine learning models.

**Override Frequency:** How often staff find it necessary to use the Auto ETA. A high frequency could mean the rules need some adjustment.

# Chapter 7

# Results and Discussion

## 7.1 Introduction and Validation Framework

This chapter presents the quantitative results derived from the performance evaluation protocols defined in Chapter 6. The primary purpose is to empirically validate the efficacy of the Hybrid Control System (HCS) implemented in the Cafeteria Helpdesk against its core objective: improving service coordination and reducing student wait times. The discussion analyzes the observed metrics across three dimensions: Service Coordination Efficiency (Queue Time Reduction), Predictive Accuracy and Reliability, and System Scalability and Real-Time Fidelity.

All results presented herein were obtained under controlled, simulated Peak Load conditions (concurrent orders exceeding), ensuring a rigorous test of the system's ability to handle operational stress.

## 7.2 Results on Service Coordination Efficiency

The most critical validation of the HCS is its ability to reduce the time students spend waiting for their order after it is ready for pickup, thereby eliminating the chaotic physical queue. This was tested by comparing the Baseline (FIFO) model against the HCS (JIT Coordination) model.

### 7.2.1 Primary Metric: Average Student Pickup Queue Time

The experiment tracked the mean time students spent waiting between the order achieving the READY status and the Actual Pickup Time.

## 7.3 Results on Predictive Accuracy and Hybrid Control

These results validate the intelligence of the system, specifically the interplay between the automated Auto ETA algorithm and the essential staff-driven Manual Override.

### 7.3.1 Predictive Accuracy: On-Time Pickup Percentage

The metric tracks the percentage of orders picked up within a tight ±3-minute tolerance of the communicated finalETA.

| Scenario | Total Orders | Orders Picked Up On-Time (±3 mins) | On-Time Pickup Percentage |
|---|---|---|---|
| HCS (Hybrid) | 100 | 92 | 92.0% |
| FIFO (Estimated) | 100 | ~50 (Based on 50% random arrival) | 50.0% |

Discussion: The high On-Time Pickup Percentage validates the reliability of the as the single source of truth. This reliability is the foundation of the system's ability to coordinate arrivals. The success is attributed directly to the Hybrid Control System, where the automated provided an accurate baseline, and the ensured necessary corrections were made instantly by kitchen staff to account for unpredictable, real-time events (e.g., equipment delay), preventing system misinformation.

### 7.3.2 Hybrid Control Analysis: Override Frequency and Offset

This analysis examines how the kitchen staff interacted with the HCS, providing feedback on the initial Auto ETA algorithm's performance.

| Metric | Result (Orders Sampled) | Implication |
|---|---|---|
| Manual Override Frequency | 18% (18 out of 100 orders) | The 18% intervention rate demonstrates that the Manual Override is necessary for a functional system, primarily during acute peak load. |
| Average Offset of Override | +4 minutes 30 seconds | When staff intervened, the system's Auto ETA was, on average, too optimistic by 4.5 minutes. This suggests the initial queue Wait calculation should be conservatively increased by 2-3 minutes in future algorithm calibration. |

Discussion:

The data confirms that the Auto ETA calculation—based on MaxPreps and queue Wait—is fundamentally sound but requires human adjustment during periods of high contention. The Hybrid Control System successfully captured the necessary human correction factor (the 4.5minute offset) within the audit log. This log is the crucial dataset required for the next phase of the project: training a Machine Learning model to automatically predict and incorporate this correction factor.

## 7.4 Results on System Reliability and Scalability

These metrics validate the performance and architectural choices of the serverless environment.

### 7.4.1 Real-Time Fidelity (Latency Measurement)

Latency was measured as the delay between the Cloud Function completing a status update and the Student App receiving the updated data via the Firestore Snapshot Listener.

| Operation | Average Latency | Target Latency | Validation Status |
|---|---|---|---|
| KDS Update | 310 milliseconds | ≤500 milliseconds | SUCCESS |

### 7.4.2 Architectural Scalability

During the peak load simulation (100 orders concurrent within 5 minutes), the system exhibited zero service degradation or failure, and the calculation time remained constant (). This validates the use of Cloud Functions and Firestore for their inherent ability to auto-scale, proving the architecture is robust and economically viable for unpredictable canteen traffic.

# Chapter 8

# Conclusion and Future Work

## 8.1 Conclusion and Project Summary

The Cafeteria Helpdesk project successfully conceived, designed, developed, and validated a novel Hybrid Control System (HCS) for real-time, JIT (Just-in-Time) pickup coordination in a high-density service environment. The system decisively addressed the core problem of temporal uncertainty and operational rigidity identified in Chapter 3, transforming the chaotic physical student queue into an efficient, managed virtual system.

The project's success is confirmed by the quantitative results derived from the experimental validation (Chapter 7):

1. **Objective Achievement (Queue Efficiency):** The system demonstrated a statistically significant reduction in the average student pickup queue time (e.g., reduction reported in Chapter 7), validating the efficacy of decoupling the physical queue from the virtual queue via dynamic ETA communication.

2. **Architectural Validation:** The selection of the Flutter/Firebase Serverless Architecture was validated by achieving high real-time fidelity (low latency on status updates) and demonstrated automatic scalability under simulated peak load conditions.

3. **Hybrid Control Validation:** The core innovation, the HCS, proved essential, achieving a high On-Time Pickup Percentage. This accuracy was sustained by the crucial integration of the automated baseline and the authoritative staff intervention via the mobile KDS, providing a reliable system that is both intelligent and pragmatic.

In summary, the Cafeteria Helpdesk represents a robust, cost-effective, and scalable engineering solution that transforms inefficient service operations into a predictable, usercentric JIT coordination engine, providing a measurable improvement in student experience and canteen throughput.

## 8.2 Project Limitations and Constraints

While the current system fulfils all defined objectives for the Minimum Viable Product (MVP), it is important to formally recognize the limitations inherent in this phase of development, which directly inform the future work strategy:

1. **Rule-Based Dependency:** The current calculation relies on a deterministic, rule-based algorithm (MaxPreps + Queue Wait). This model cannot account for non-linear, stochastic variables that influence real-world service time, such as staff experience, time-of-day variability, or temporary equipment degradation. This requires manual correction by staff (the override offset).

2. **Limited Staff Specialization:** The current KDS model assumes a generalized cooking process. It does not yet account for complex kitchen workflows involving specialized stations (e.g., separate Fryer Station, Cold Prep Station, Drink Station), which would require advanced parallel processing logic.

3. **Absence of Predictive Feedback Loop:** Although the system collects rich data (the vs. offset), it currently lacks the capability to learn from this data to refine its prediction, operating as a reactive system rather than a fully proactive one.

# Chapter 9

# Future Scope

## 9.Strategic Future Work and Academic Extension

The foundation established by the HCS architecture provides a clear and fertile ground for strategic academic extension and commercial application. The following outlines the key recommendations for future phases of development:

### 9.1 Phase I: Advanced Predictive Modelling (Machine Learning Integration)

The primary and most significant future work involves evolving the from a rule-based calculation to a data-driven predictive model.

- **Objective:** Develop and deploy a Machine Learning (ML) model to forecast service time.
- **Methodology:** Utilize the rich, auditable data set captured by the HCS (Auto ETA, Manual ETA, Actual Pickup Time, Item Composition) to train a Regression Model (e.g., Gradient Boosting or Neural Network).
- **Key Features for ML Model:** Inputs would include Temporal Variables (Time of Day, Day of Week), Item Complexity Indicators (), and the historical Kitchen Load Index.
- **Deployment:** The trained ML model would be deployed via Cloud Functions or Vertex AI and replace the current static algorithm, providing a genuinely dynamic, self-learning prediction that requires minimal staff correction.

### 9.2 Phase II: Operational Optimization and Parallel Processing

To enhance kitchen efficiency, the system needs to incorporate fine-grained operational intelligence:

- **Specialized Station Routing:** Modify the KDS to support multiple kitchen terminals, allowing orders to be automatically routed to specific Preparation Stations (e.g., Sandwich KDS, Hot Drinks KDS). This requires updating the data model to include item routing flags.
- **Optimal Order Sequencing:** Implement a system to suggest the optimal order sequence for staff. Instead of strictly chronological sorting, the KDS could suggest prioritizing short orders that finish quickly (to increase throughput) or grouping similar long-prep items (to optimize resource utilization).

### 9.3 Phase III: User Experience and Commercial Features

To transition the system toward commercial viability, several user-facing features are recommended:

- **Payment Gateway Integration:** Implement secure payment processing (Razor pay) via a dedicated Cloud Function to transition from a status tracker to a full Point-of-Sale (POS) system.
- **Order Customization Interface:** Develop a modular UI allowing students to modify order items (e.g., add-ons, modifiers) which must be securely mapped to update the underlying metric, feeding back into the ETA algorithm.
- **Rating and Feedback System:** Integrate a simple rating system to gather user experience data on accuracy and food quality, completing the service operations feedback loop.

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